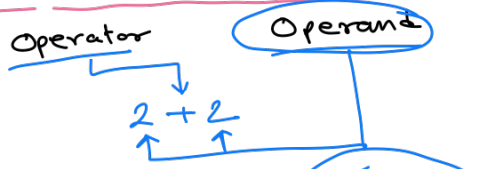


Dart Programming Language



- Unary
- ✓ x++
- y
- 1) Unary Plus (+)
 - 2) Unary Minus (-)
 - 3) Unary Increment
 - Prefix: ++x
 - Postfix: x++
 - 4) Decrementor
 - Prefix: --x
 - Postfix: x--
 - 5) Logical NOT (!)
 - 6) Bitwise NOT (~)
 - 7) (await)
- Binary
- x <= y
- 1) Arithmetic op
+ - * / %
 - 2) Assignment op
= += -= *= /= %=
/= %=
 - 3) Logical op.
& & ||
 - 4) Relational op
==, !=, >=, <=, >, <
 - 5) Bitwise op
& | ^ << >>
 - 6) Null aware
?? ??= ?
 - 7) [] []=
set set
 - 8) .
 - 9) is is! as
- Unary
- Cond? ex: x < 2 ? 1 : 0
- Conditional op.

Special

Spread ...

...?

Cascade

...

```
void main() {
  int x = 3;
  int y = x;
  print(y); // 3
}
```

```
void main() {
  int x = 4;
  int y = --x;
  print(y); // 3
}
```

Increment

```
void main() {
  int x = 3;
  int y = 3;
  print(x++); // 3
  print(++y); // 4
}
```

Decrement

```
void main() {
  int x = 5;
  int y = 5;
  print(--x); // 4
  print(y--); // 5
}
```

Precedence

1) Unary postfix $x++, y--$
 $()[], \dots$
 $?. !$

2) Unary prefix $--x, ++y$
 $await,$
 $\sim x$
 $-x$

3) Multiplicative $*$ $/$
 $\% \quad \backslash$

4) Additive $+$ $-$

5) Shift $<<$ $>>$

6) Bitwise AND $\&$

7) Bitwise XOR $\wedge$

8) Bitwise OR $|$

9) Relational
 Type test/type
 cast $>=$ $>$
 $<=$ $<$
 as, is
 $is!$

10) equality $==$ $!=$

11) logical AND $\&\&$

12) logical OR $||$

13) if null $??$

14) Conditional $cond ? ex1 : ex2;$

15) cascade $\dots ? \dots$

16) assignment $=$ $*=$ $/=$ $+=$
 $-=$

17) Spread $\dots \dots ?$

Associativity

NONE

NONE

$\longrightarrow$ L-R

$\longrightarrow$ L-R

$\longrightarrow$ L-R

$\longrightarrow$ L-R

$\longrightarrow$ L-R

$\longrightarrow$ L-R

$\longrightarrow$ NONE

$\longrightarrow$ NONE

$\longrightarrow$ L-R

$\longrightarrow$ L-R

$\longrightarrow$ L-R

$\longrightarrow$ R-L

$\longrightarrow$ L-R

$\longrightarrow$ R-L

$\longrightarrow$ NONE

$$\underline{5 \times 3 - 2 \div 2} \quad (L-R)$$

$$= 15 - 1$$

$$= 14$$

Print $(5 > 3 \&\& 2 < 4)$
 $\downarrow$ $\downarrow$ $\downarrow$
 R.O.P L.O.P R.O.P
 (true & true)
true

Print $(a \& b | c)$

$a = 7 \rightarrow 111$
 $b = 3 \rightarrow 11$
 $c = 4 \rightarrow 100$

$$\begin{array}{r} 111 \\ \& 011 \\ \hline 011 \\ \hline 1100 \\ \hline 111 \rightarrow 7 \end{array}$$

Print $(x < 2 \& y)$

$x = 4 \rightarrow 100$
 $y = 2 \rightarrow 10$

$$\begin{array}{r} 100 \\ 10000 \\ \& 00010 \\ \hline 00000 \end{array} \Rightarrow 0$$